

Exploratory Data Analysis (EDA) Report: Analysis of Digital Marketing Channels and Their Impact on Sales

Introduction

Exploratory Data Analysis (EDA) is a crucial step in understanding data patterns, detecting anomalies, and preparing data for machine learning models. In this project, EDA is performed on a marketing dataset to analyze customer behavior and predict total purchases based on search, advertisement, and social media activities.

Project Objective

The primary objective of this project is to analyze marketing campaign data and predict total customer purchases. The project also classifies marketing performance into Hit or Flop categories and visualizes trends using statistical and graphical methods.

Dataset Description

Dataset Name: Marketing.csv

The dataset contains weekly marketing data collected from multiple channels such as search engines, advertisements, and social media platforms.

Variables in the Dataset

Numerical Variables:

- Search Sessions
- Ad Clicks
- Social Views
- Ad Budget
- Search Purchases
- Ad Purchases
- Social Purchases

Derived Variables:

- Total Purchases
 - Performance (Hit / Flop)
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Project Structure

1. Understand the Problem and Data
 2. Import and Inspect the Data
 3. Data Cleaning and Preprocessing
 4. Statistical Analysis
 5. Data Visualization
 6. Correlation Analysis
 7. Hit / Flop Classification
 8. Clustering Model
 9. Model Training
 10. Dashboard
 11. Conclusion
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1. Import and Inspect the Data

The dataset is loaded using Pandas through the `data_loader.py` file. Initial inspection is performed using `head()` and `info()` functions to understand the structure, column names, and data types.

```
Unnamed: 0  week_number  search_sessions  ...  audience_male_percent  ad_budget  category
0           1           1           892  ...           68.8       13430.0      Base
1           2           2          1271  ...           67.6       14999.0      Base
2           3           3           976  ...           66.5       14126.0      Base
3           4           4          1394  ...           65.4       10666.0      Base
4           5           5          1391  ...           64.3       15418.0      Base

[5 rows x 13 columns]
```

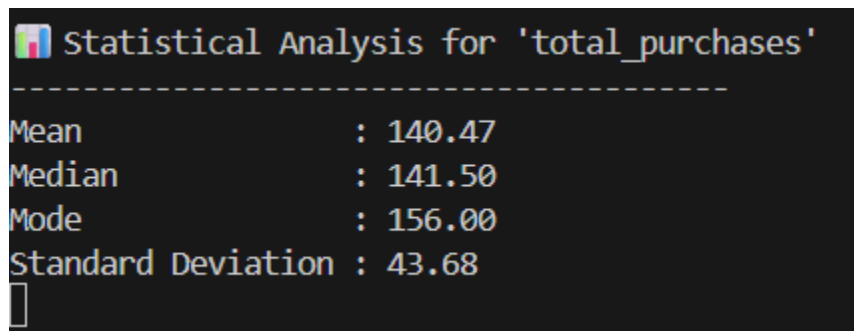
```
Data columns (total 13 columns):
#      Column                                Non-Null Count  Dtype
---  -
0      Unnamed: 0                             156 non-null   int64
1      week_number                           156 non-null   int64
2      search_sessions                       156 non-null   int64
3      search_purchases                      156 non-null   int64
4      ad_clicks                             156 non-null   int64
5      ad_purchases                          156 non-null   int64
6      social_views                          156 non-null   int64
7      social_clicks                         156 non-null   int64
8      social_purchases                      156 non-null   int64
9      audience_avg_age                     156 non-null   float64
10     audience_male_percent                 156 non-null   float64
11     ad_budget                             156 non-null   float64
12     category                             156 non-null   object
dtypes: float64(3), int64(9), object(1)
memory usage: 16.0+ KB
None
```

2. Data Cleaning and Preprocessing

Data cleaning is performed using `data_cleaning.py`. Missing values are handled using forward fill techniques. Unnecessary columns are removed, and numeric columns are converted safely. A new feature called `total_purchases` is created by summing purchases from all channels.

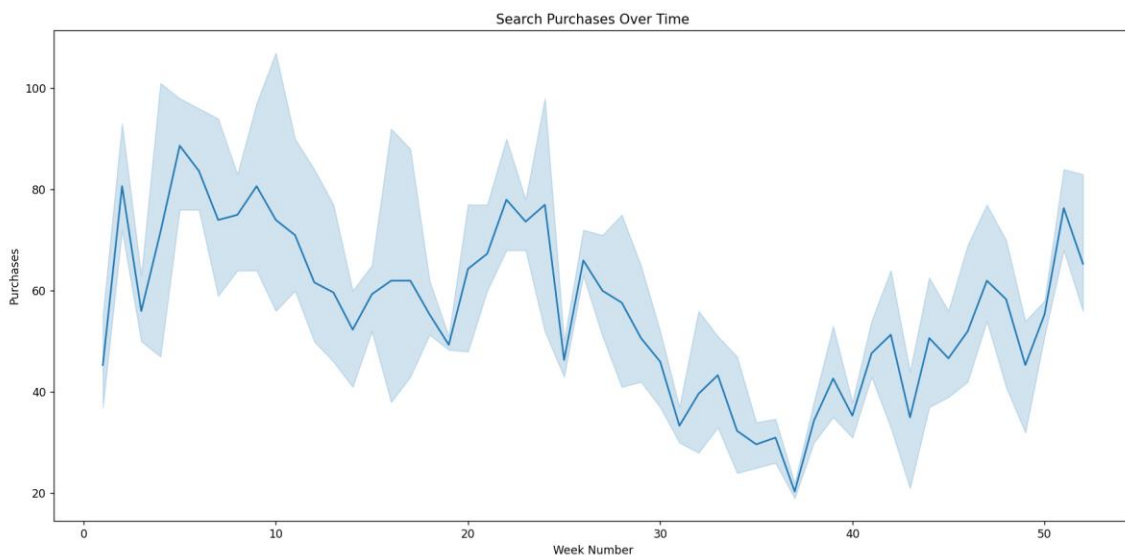
3. Statistical Analysis

Statistical measures such as Mean, Median, Mode, and Standard Deviation are calculated for total purchases. These metrics provide insights into customer purchase behavior and variability.



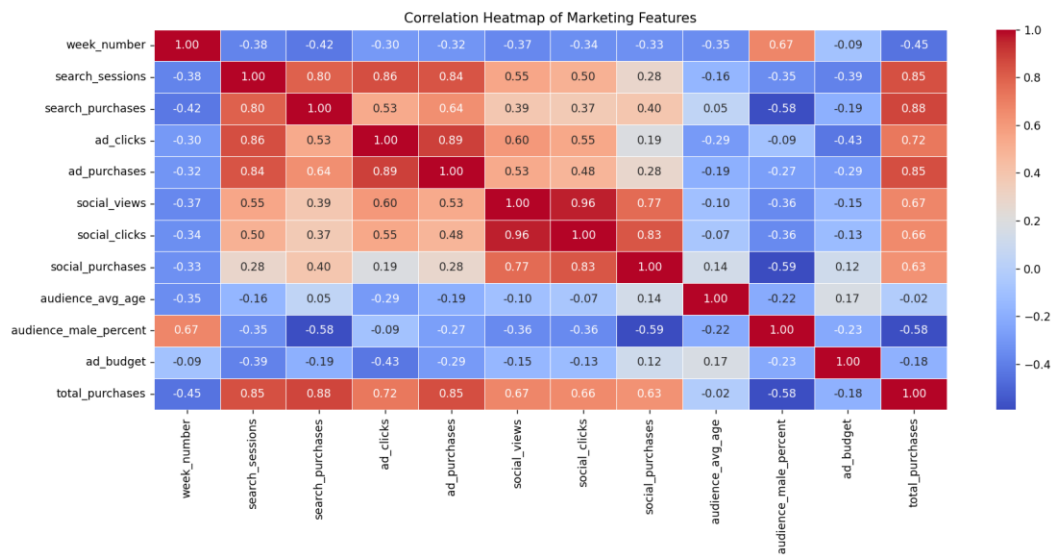
4. Data Visualization

Visualizations are created using Matplotlib and Seaborn. Line plots show weekly purchase trends, and heatmaps illustrate correlations between numerical variables.



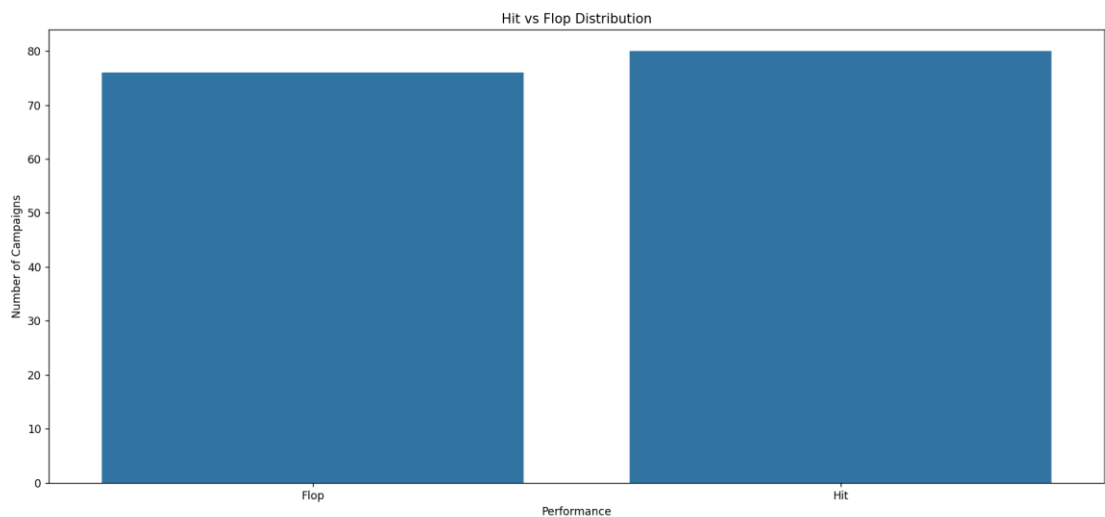
5. Correlation Analysis

A correlation heatmap is generated to identify relationships among numerical features. Strong correlations help determine which variables influence total purchases.



6. Hit / Flop Classification

Marketing performance is classified as Hit or Flop based on average total purchases. If total purchases are greater than or equal to the average, the performance is labeled as Hit; otherwise, it is labeled as Flop.

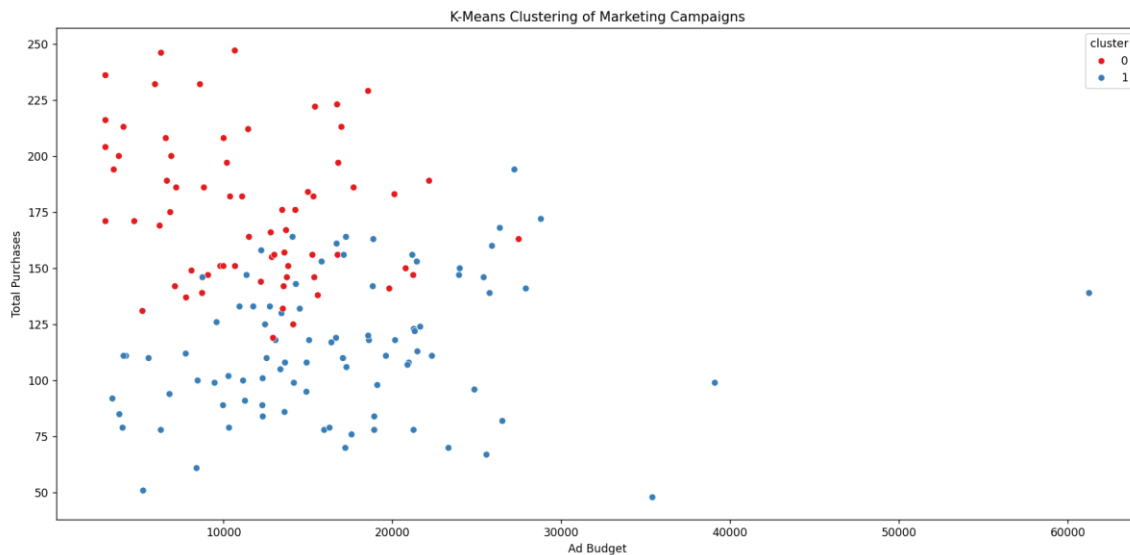


7. Clustering Analysis

File Used: [clustering.py](#)

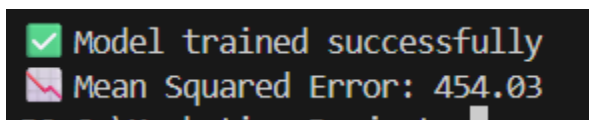
K-Means clustering was applied with $k = 3$ to segment games into performance-based clusters. This unsupervised approach reveals hidden groupings among games with similar characteristics.

Kmean clustering:



8. Model Training

A Linear Regression model is trained using features such as search sessions, ad clicks, social views, and ad budget. Model performance is evaluated using Mean Squared Error (MSE).

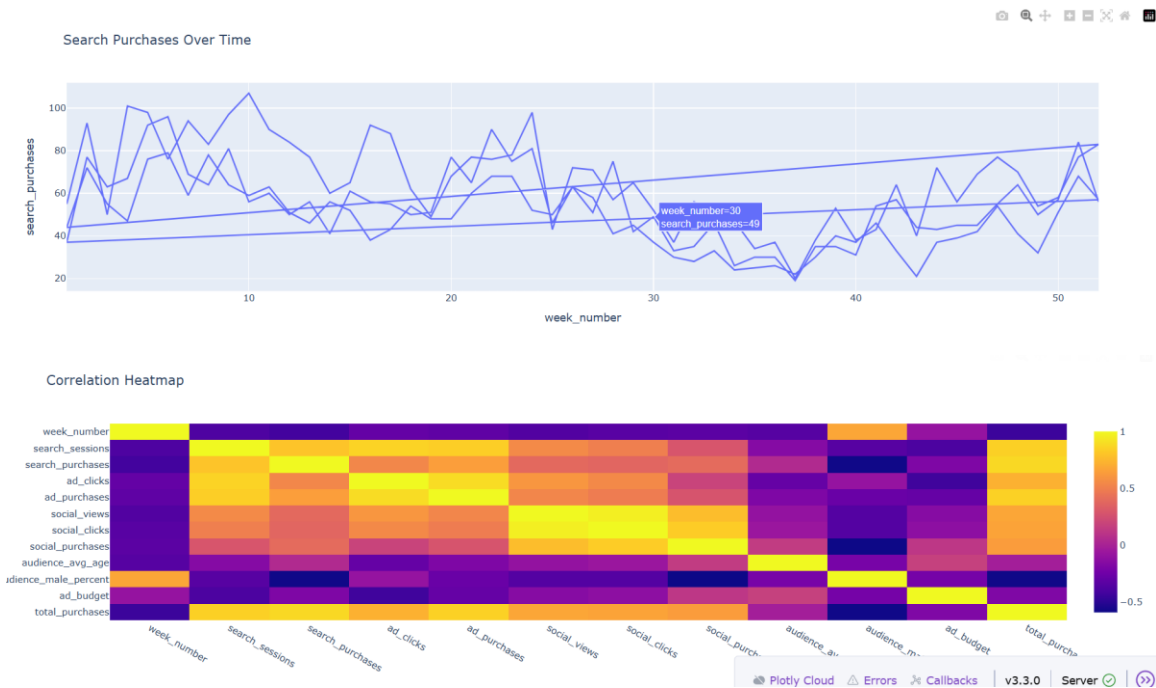


9. Dashboard

An interactive dashboard is developed using Dash and Plotly. It displays EDA visualizations and model performance metrics in a user-friendly interface.

Marketing Dashboard

Model MSE: 454.03



Conclusion

This project demonstrates an end-to-end marketing data analysis workflow. Through data cleaning, EDA, statistical analysis, visualization, and machine learning, valuable insights into customer purchase behavior are obtained. The results help businesses optimize marketing strategies and improve decision-making.